

Peer-reviewed journal articles and refereed conference proceedings

- Craig, P.**, Roy, A., Varshney, N., Asadizanjani, N. (2024). Advancing PCB assurance towards netlist extraction with the integration of X-ray imaging and semi-supervised learning techniques. 2024 IEEE Research and Applications of Photonics in Defense Conference (RAPID), pp. 1–2.
- Varshney, N., Ghosh, S., **Craig, P.**, Kottur, H. R., Dalir, H., Asadizanjani, N. (2024). Challenges and opportunities in non-destructive characterization of stacked IC packaging: Insights from SAM and 3D X-ray analysis. *Developments in X-Ray Tomography XV*, SPIE Vol. 13152, pp. 415–422.
- Craig, P.**, Xi, C., Varshney, N., Mitchell, W., Ghosh, S., Asadizanjani, N. (2024). Terahertz time-domain spectroscopy (THz-TDS) fingerprinting for integrated circuit (IC) identification in tracking and tracing applications. *Terahertz Emitters, Receivers, and Applications XV*, SPIE Vol. 13141, pp. 52–57.
- Craig, P.**, Varshney, N., Roy, A., Woychik, C., Dalir, H., Asadizanjani, N. (2024). Motivating automated multimodal failure analysis for heterogeneously integrated devices. *Developments in X-Ray Tomography XV*, SPIE Vol. 13152, pp. 423–428.
- Craig, P.**, Varshney, N., Roy, A., Ghosh, S., Patil, C., Dalir, H., Asadizanjani, N. (2024). Multi-modal printed circuit board netlist extraction with X-ray and optical imaging. *Developments in X-Ray Tomography XV*, SPIE Vol. 13152, pp. 88–92.
- Ghosh, S., Varshney, N., Al Hasan, M. M., Roy, A., **Craig, P.**, Koppal, S. J., Dalir, H., Asadizanjani, N. (2024). Exploring physics-informed machine learning for system matrix formulation in X-ray imaging forward models. *Developments in X-Ray Tomography XV*, SPIE Vol. 13152, pp. 377–384.
- Craig, P.**, Pearson, J., Ghosh, S., Varshney, N., Koppal, S. J., Asadizanjani, N. (2024). Optical automated interconnect inspection of printed circuit boards. *International Symposium for Testing and Failure Analysis (ISTFA)*, Vol. 84918, pp. 22–27. ASM International.
- Roy, A., Varshney, N., **Craig, P.**, Ghosh, S., Al Hasan, M. M., Asadizanjani, N. (2025). SegPCBX: Redefining automated PCB inspection with a novel PCB X-ray dataset for component analysis. *Metrology, Inspection, and Process Control XXXIX*, SPIE Vol. 13426, pp. 856–864.
- Yahyaee, K., Khan, M. S. M., Ghosh, S., Roy, A., **Craig, P.**, Varshney, N., Asadizanjani, N. (2025). X-ADAPT: AI-driven design-based strategy to address X-ray compatibility challenges in advanced packaging metrology. *Metrology, Inspection, and Process Control XXXIX*, SPIE Vol. 13426, pp. 865–875.
- Roy, A., Ghosh, S., **Craig, P.**, Varshney, N., Al Hasan, M. M., Asadizanjani, N. (2025). Preserving fine details in PCB X-ray imaging: A patch-based data augmentation method for improved segmentation and detection. *Metrology, Inspection, and Process Control XXXIX*, SPIE Vol. 13426, pp. 845–855.
- Biswas, L. K., **Craig, P. J.**, Varshney, N., Roy, A., Godinez, C. D., Paki, P., Asadizanjani, N. (2025). Advanced packaging raw materials assurance by characterization. *Metrology, Inspection, and Process Control XXXIX*, SPIE Vol. 13426, pp. 297–303.
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Shovon, M. S. I., Ghosh, S., **Craig, P. J.**, Huang, C.-C., Pai, C.-Y., Asadizanjani, N. (2025). A novel framework for identifying counterfeit ICs via pinhole evaluation. 2025 IEEE Research and Applications of Photonics in Defense Conference (RAPID), pp. 1–2.

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Noor, R., Kottur, H. R., **Craig, P. J.**, Biswas, L. K., Khan, M. S. M., Varshney, N., Dalir, H., Akçalı, E., Ghane Motlagh, B., Woychik, C., et al. (2023). U.S. microelectronics packaging ecosystem: Challenges and opportunities. arXiv preprint arXiv:2310.11651.

Patents

Asadi-Zanjani, N., Varshney, N., Ghosh, S., Roy, A., **Craig, P.**, Al Hasan, M. M. (2026, January 15). Explainable AI metrics for IC packaging inspection. U.S. Patent Application No. 19/263,036.

Asadi-Zanjani, N., Varshney, N., Ghosh, S., Roy, A., **Craig, P.**, Al Hasan, M. M. (2026, January 29). Image reconstruction with multimodal fusion and physics-informed neural network. U.S. Patent Application No. 19/275,418.